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DETERMINATION OF THE THERMOPHYSICAL CHARACTERISTICS OF LIQUIDS BY THE LASER-BURST METHOD

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From the results of mathematical modeling of heat-conduction processes under conditions corresponding to experiments on determination of the thermophysical characteristics of liquids by the laser-burst method, the authors have substantiated possible ranges of variation in basic factors. The authors have established the values of the heat fluxes and of the thicknesses of liquid layers that ensure minimum errors in determining the thermal diffusivity of the liquid when the surface of a measurement cell is exposed to a laser burst of finite duration.

Keywords: *laser-burst method, thermal diffusivity of the liquid, layer thickness, heat flux, methodological errors, heat-conduction equations, numerical solution.*

Introduction. One efficient method widely used to determine thermophysical characteristics of materials in a moderate temperature range is the laser-burst (laser-pulse) method [1–3], which is based on the absorption, in a thin layer of the frontal "hot" surface of a sample of thickness L, of a radiant-energy pulse, and also on the approximate solution of a one-dimensional heat-conduction problem for an infinite plate. From the obtained temporal thermogram of the sample′s surface, which is opposite to the heated one, we determine the maximum superheating temperature $T_{\max}$ and the time of reaching half the maximum superheating temperature $\tau_{0.5}$. The specific heat c, the thermal diffusivity a, and the thermal conductivity λ of the material are calculated from the expressions [1]

$$c = Q/(T_{\max}\rho L)\ , \quad a = 0.1388L^2/\tau_{0.5}\ , \quad \lambda = ac\rho\ . \tag{1}$$

The laser-burst method [1–3] is frequently employed in determining the thermophysical characteristics of materials in a solid state. In recent years, attention has also been drawn to the use of this method for determining the thermal diffusivity of liquids and of emulsions based on them [4–8]. When the thermophysical characteristics of low-thermal-conductivity liquids are determined by the laser-burst method [1–3] the "sample" of the substance under study must be in a specially created measurement cell ensuring intense heating of a homogeneous and plane-parallel liquid layer of certain thickness.

In determining the thermophysical characteristics of liquid media, researchers [4–8] run into substantial constraints imposed on the ranges of operating temperatures (they must be lower than the boiling point of the liquids T_b). In connection with these constraints on the intensity of heating (heat flux q), the temperature of the investigated liquid must be moderate. However, the latter constraint leads to a corresponding reduction in the temperature of the "cold" (opposite to the heated one) surface $T_{\max}$ of the medium (or of the measurement cell): in the basic characteristic of the heat-transfer processes that is used in the laser-burst method [1–3]. With the aim of raising $T_{\max}$ as much as possible under the considered complicated conditions, we can decrease the thickness of the liquid layer Δ to a value minimum acceptable in practice.

However, selection of q, $T_{\max}$, and Δ values sufficient to obtain reliable values of thermophysical characteristics of a low-thermal-conductivity liquid is a complex polyvariant problem whose solution through experimental investigations is very labor-intensive and time-consuming in each individual case of a concrete liquid. It is of (both theoretical and practical) interest to develop a predictive model of the process in question that connects the error in determining a concrete thermophysical characteristic (e.g., the thermal diffusivity of a liquid a) to the parameters q and Δ with constraints on the maximum values of the medium′s superheating temperature (boiling point) T_b and at minimum acceptable values of $T_{\max}$.

This problem can be solved using the earlier developed approach [9, 10] to assessment of the errors of the thermal diffusivity, thermal conductivity, and specific heat of materials in a solid state. However, in determining the thermophysical

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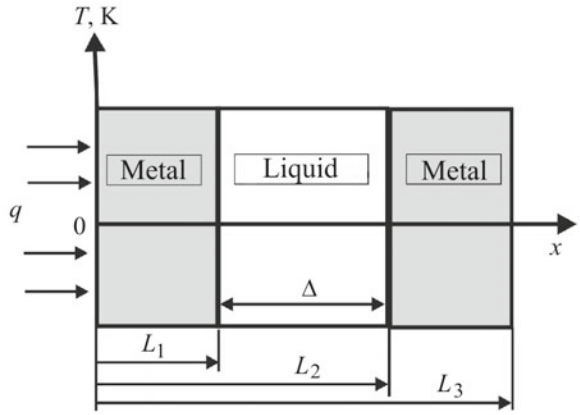


Fig. 1. Diagram of the solution domain.

characteristics of liquids, the latter must be under conditions ensuring, on the one hand, reliable measurement of T_{max} [1–3] of the "cold surface," and on the other, elimination of the conditions of rise of the heating-surface temperature above T_b. This experiment is possible if the liquid is distributed as a fairly thin layer between two plates (Fig. 1) fabricated from a high-thermal-conductivity metal (e.g., copper, aluminum, and steel) or alloy. In the first step of such analysis, the problem of selection of the laser-burst heat flux eliminating the rise of the heating-surface temperature above the boiling point of the liquid T_b is relevant. In the second step, selection of the layer thickness of the concrete liquid under study, which ensures reliable determination of the thermophysical characteristics by the laser-burst method [1], is a relevant problem.

The problem is that Parker et al. [1] have proposed a mathematical expression to determine thermophysical characteristics from the results of a simplified solution of the heat-conduction problem for a single-layer plate under the conditions of pulsed heating of the latter. When liquids are investigated, the "sample" of the substance under study is a three-layer system. Thin sheets (foil) of a high-thermal-conductivity metal (or alloy) may be outer layers, and the liquid under study, the inner layer. Heat transfer in a three-layer system with different thermophysical characteristics of the materials of the layers generally occurs according to a more complex scheme, even if the thermal conductivity of each layer differs only slightly. In the case in question, the values of the thermophysical characteristics of the metals and liquids in the system (Fig. 1) may differ hundreds of times or more, which can produce great errors in determining c, a, and λ.

The present work seeks to substantiate, from the results of mathematical modeling, conditions for the experiments on determination of the thermal diffusivity of typical organic liquids, i.e., ethanol and toluene, by the laser-burst method [1–3].

Formulation of the Problem. In formulating the problem, we assumed that the thermophysical characteristics of the materials of a measurement cell (metal) and a liquid (ethanol and toluene) are temperature-independent, which has been substantiated for many materials in a real narrow range of possible temperature variation in the process of the experiment (even at high values of the specific heat fluxes to the sample′s heated surface) in connection with the short pulse duration.

Under the assumption that the transverse dimension of the measurement cell is many times larger than the thickness of the liquid layer, we disregarded a possible two-dimensional heat transfer. To assess the scope of influence of the heat flux of the pulse and of the layer thickness of the investigated liquid on errors in determining its thermal diffusivity, we solved the one-dimensional heat-conduction problem for a three-layer inhomogeneous unbounded plate when its surface is exposed to collimated laser radiation of short (1 ms) time duration τ_{pul} under the assumption of ideal thermal contact on the boundaries between the layers.

The problem was solved in a Cartesian coordinate system. The solution domain represented a three-layer inhomogeneous unbounded plate of total thickness L_3 (Fig. 1) a portion of whose boundary $x = 0$ was heated by an energy flux perpendicularly to the plate of the heated surface at the radiation density q and the pulse duration τ_{pul}.

Mathematical Model and Solution Method. A mathematical model corresponding to the adopted formulation of the problem (Fig. 1) includes a system of three nonsteady partial differential heat-conduction equations with boundary and initial conditions

$$c_1\rho_1 \frac{\partial T_1}{\partial t} = \lambda_1 \left(\frac{\partial^2 T_1}{\partial x^2} \right), \quad 0 < x < L_1 ; \tag{2}$$

$$c_2\rho_2 \frac{\partial T_2}{\partial t} = \lambda_2 \left(\frac{\partial^2 T_2}{\partial x^2} \right), \quad L_1 < x < L_2 ; \tag{3}$$

$$c_3\rho_3 \frac{\partial T_3}{\partial t} = \lambda_3 \left(\frac{\partial^2 T_3}{\partial x^2} \right), \quad L_2 < x < L_3 ; \tag{4}$$

$$x = 0 : \quad -\lambda_1 \frac{\partial T_1}{\partial x} = q , \quad \tau_{\text{pul}} > t > 0 ; \tag{5}$$

$$x = 0 : \quad -\lambda_1 \frac{\partial T_1}{\partial x} = 0 , \quad t > \tau_{\text{pul}} ; \tag{6}$$

$$x = L_1 : \quad T_1 = T_2 , \quad -\lambda_1 \left(\frac{\partial T_1}{\partial x} \right) = -\lambda_2 \left(\frac{\partial T_2}{\partial x} \right); \tag{7}$$

$$x = L_2 : \quad T_2 = T_3 , \quad -\lambda_2 \left(\frac{\partial T_2}{\partial x} \right) = -\lambda_3 \left(\frac{\partial T_3}{\partial x} \right); \tag{8}$$

$$x = L_3 : \quad \lambda_3 \frac{\partial T_3}{\partial x} = 0, \quad t > 0 ; \tag{9}$$

$$t = 0 : \quad T_1 = T_2 = T_3 = T_0 . \tag{10}$$

Boundary-value problem (2)–(10) has been solved by the finite-difference method [11]. In the considered case of the three-layer plate the problem is not only singularly perturbed [12], as in [9, 10], but it is also has transfer coefficients undergoing a discontinuity on the boundaries between the media. In solving the boundary-value problem of mathematical physics, it is necessary to use special analytical tools developed with the aim of modeling mathematically fast heat- and mass-transfer processes and intense physicochemical transformations in heatproof materials [13] and combustible liquids [14] or for solution of problems of conjugate heat transfer in closed gas cavities with local heat-release and heat-accumulating solid walls [15].

The heat fluxes and the pulse durations were selected in accordance with the characteristics of today′s laser facilities on determination of thermal diffusivity and thermal conductivity by the laser-burst method. During the numerical simulation, we took the following parameters of the laser heating source: energy of the laser pulse to 25 J/pulse, pulse duration τ_{pul} = 0.3–1 ms, and radiation wavelength 1.06 μm [8].

Results and Discussion. It has been established [9, 10] that in determining the thermophysical characteristics of solid materials by the method of pulse laser action, there can be methodological errors due to the assumptions made during the formulation of the problem [1–3]. Errors in determining the thermal diffusivity a of materials by the laser-burst method were established [9, 10] by comparison of the reference values of a and of those resulting from the numeral simulation of the conditions of implementation of experiments on determination of the thermal diffusivity by the laser-burst method [1–3]. In the same manner as in [9, 10], we determined errors of the thermal diffusivity of the liquids in the present work. Verification of the mathematical model and assessment of the reliability of results of numeral investigation allowed their comparison with the existing experimental results [16], and also a check of the conservatism of the used difference scheme analogously to [14, 15].

Numerical simulation of the process of heat transfer in liquids was carried out with the example of ethanol (thermophysical characteristics λ = 0.172 W/(m·K), c = 1909 J/(kg·K), and ρ = 812.1 kg/m^3) and toluene (λ = 0.138 W/(m·K), c = 1630 J/(kg·K), and ρ = 885.5 kg/m^3) [17]. The initial temperature of the liquids was T_0 = 273 K at pressures lower than those on the saturation line P_{sat}. We took P_{sat} = 16 mm Hg for ethanol and P_{sat} = 6.5 mm Hg for toluene. The measurement cell was manufactured from steel with a wall thickness of 0.5 mm (λ = 20 W/(m·K), c = 551 J/(kg·K), and ρ = 7894 kg/m^3) [18] at the values of the heat flux $q = 10^7$–10^8 W/m^2 and a pulse duration of 1.0 ms. The taken value of the initial temperature was T_0 = 273 K (characteristic of the experiments in question). The thickness of the liquid layer varied from 0.8 to 1.2 mm. The time step of the difference grid was as small as 10^{-6} s. The initial coordinate step was $x = 5 \cdot 10^{-7}$ m, and near the boundaries

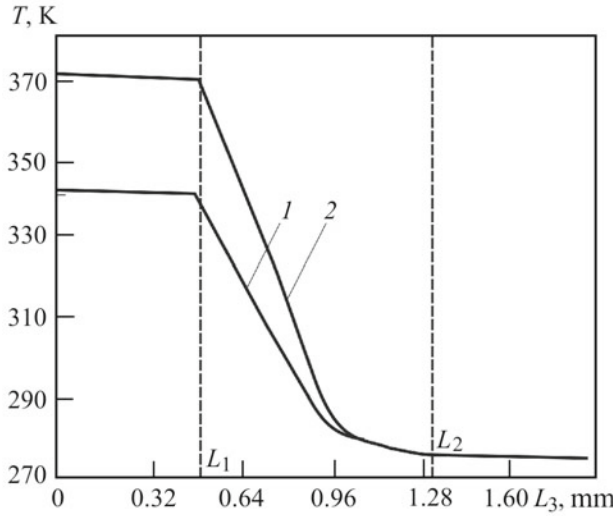


Fig. 2. Distribution of $T(x)$ in the measurement cell at an instant of time of 0.1 s: 1) ethanol ($q = 5.5 \cdot 10^7$ W/m^2); 2) toluene ($q = 7.3 \cdot 10^7$ W/m^2).

of the materials, by analogy with [15], it was selected nonuniform (according to the algorithm of geometric progression) [19]: $x_{i+1} = x_i \nu$, where ν is the coefficient of variation. As the boundary between the media was approached, the step decreased, and after crossing the boundary, it increased. The coefficient of variation ν was assumed constant and equal to 0.02.

Figure 2 gives the temperature fields in three-layer systems of thickness 1.8 mm for ethanol and toluene before reaching the conditions of boiling of the liquids. The figure clearly illustrates the possibility of determining the thermal diffusivity of the liquids by the laser-burst method. At a thickness of the walls of the steel measurement cell of 0.5 mm, the temperature of the liquid on the boundary $x = L_1$ differs only slightly from the surface temperature of the measurement cell $x = 0$ heated by the laser pulse. A slight difference in temperatures is also established on the boundaries $x = L_2$ (liquid) and $x = L_3$ ("cold" surface of the measurement cell), where T_{max}, i.e., the basic parameter of the method, is recorded [1].

A certain drawback of the laser-burst method [1–3] is the assumption that the value of the laser-radiation heat flux to the heated surface is infinitely high. In actual practice, this quantity cannot be infinitely high.

On the one hand, a heat flux of 10^7–10^8 W/m^2 may lead to a temperature growth sufficient for the liquid to start to boil, which will substantially increase the error in determining the thermal diffusivity (e.g., the boiling point T_b of ethanol is 351.4 K and of toluene, 383.6 K [18]). On the other hand, a low value of the heat flux may lead to a violation of the basic proposition of the method [1]: obtaining a "temperature superheating" T_{max}^{min} on the sample′s "cold" surface of no less than 3–6 K. Accordingly, one problem of numerical simulation was to determine maximum permissible values of the heat flux in the experiment on the "measurement cell–liquid" boundary. The performed numerical analysis has shown the necessity of decreasing substantially the limiting (maximum) heat fluxes to determine the thermal diffusivity of liquids with a low thermal conductivity by the pulse method. Thus, e.g., at thicknesses of measurement-cell walls of 0.5 mm and of the liquid layer of 0.8 mm and at a duration of the laser-heating pulse of 1.0 ms, the value of the maximum permissible heat flux was $7.38 \cdot 10^7$ W/m^2 for toluene and $5.56 \cdot 10^7$ W/m^2 for ethanol.

As a result of the numerical simulation, we have also established lower bounds of variation in the values of the heat fluxes and the thicknesses of the liquids Δ in the experiment, at which the value of T_{max}^{min} on the "cold" surface of the measurement cell corresponds to the lower limit of the permissible temperature range according to [1].

Figure 3 gives results of numerical simulation of the dependence of the temperature T_{max} on the "cold" surface of the measurement cell on the thickness of the liquid (ethanol and toluene) layer Δ with variation in the heat flux q. It has been established that the lower bounds of q and Δ in the experiments are (Fig. 3) $q = 0.4 \cdot 10^7$ W/m^2 and Δ = 0.9 mm for ethanol and $q = 0.5 \cdot 10^7$ W/m^2 and Δ = 1.1 mm for toluene.

In conducting the experiment by the laser-burst method, we must also select the thickness of the sample of the investigated material: the most efficient characteristics according to the test of validity. Results of investigations of solid materials [9] have shown a substantial influence of this factor on the errors in determining thermal diffusivity. The dependences of this error on the layer thickness Δ were also analyzed when the thermal diffusivities of the liquids were determined.

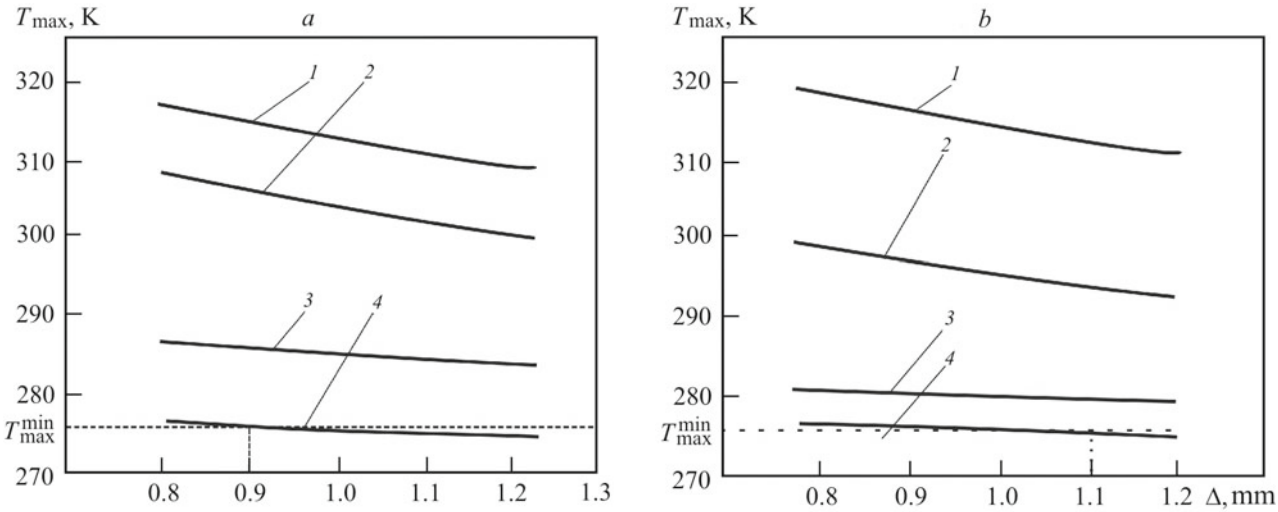


Fig. 3. Temperatures T_{max} on the "cold" surface of the measurement cell vs. thickness of the ethanol (a) and toluene (b) layer with variation in the value of the heat flux: 1) $q = 5.5 \cdot 10^7$, 2) $3.0 \cdot 10^7$, 3) $1.0 \cdot 10^7$, and 4) $0.4 \cdot 10^7$ W/m^2.

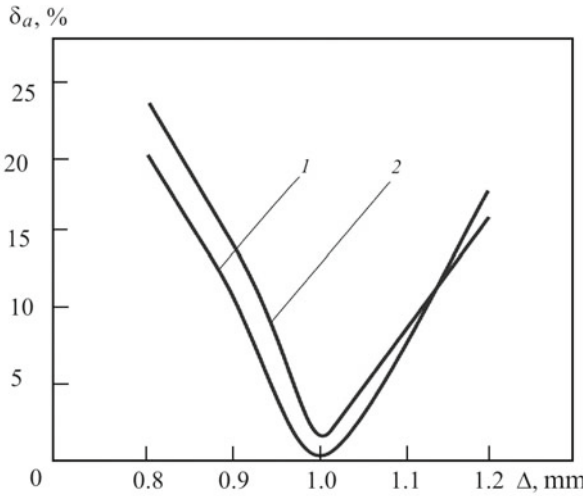


Fig. 4. Thermal-diffusivity errors δ_a vs. thickness of the liquid layer Δ: 1) toluene; 2) ethanol.

Figure 4 shows the dependences of the thermal-diffusivity errors δ_a for ethanol and toluene on the liquid-layer thickness Δ. The values of δ_a (Fig. 4) change nonmonotonically and are substantially dependent on Δ. With increase in Δ from 0.8 to 1.0 mm, the values of δ_a become lower and reach a minimum value of about 0.6% for toluene and of 2% for ethanol at Δ = 1.0 mm. With further increase in Δ, the values of δ_a grow (Fig. 4) and amount to as much as 18.6% for toluene and 13.7% for ethanol respectively at Δ = 1.2 mm. The significant variation in the quantity δ_a is due to the low thermal conductivity of the liquids under study. With increase in the thickness of the liquid layer Δ, the values of the temperature on the "cold" boundary of the measurement cell T_{max} decrease, which finally produces great errors.

The obtained results (Fig. 4) show the optimum thickness of the measurement cell in determining thermophysical characteristics of quite typical organic liquids. Great errors in determining their thermal diffusivity are due to the fact that during the pulse time (it is always finite in the experiments), the liquid is warmed up to a certain depth. The heat accumulated in this warmed-up layer ensures the warmup of the remaining ("cold") layers of the liquid. The larger the warmup depth during the pulse time and the smaller the dimension of the measurement cell, the more significant the difference between the experimental conditions and the initial Parker problem [1]. Accordingly, the smaller the cell size, the greater the methodological

error due to the finite time of action of the pulse and to the formation of the thickness of the warmed-up liquid layer during this time. At large dimensions of the measurement cell, the error in determining a is also great, since the thermal wave does not manage to exert a significant influence on the "cold" surface of the measurement sample over the pulse period. Surface temperatures are low; accordingly, the error in determining the liquid′s thermal diffusivity increases with Δ, too (Fig. 4). In this connection, a dimension of the measurement cell of 0.5 mm is most likely not optimum. As follows from the results of the performed investigations, this thickness is 1 mm.

Conclusions. On the basis of the results of the performed numerical investigations, it may be inferred that the laser-burst method can quite efficiently be employed in determining the thermal diffusivity of low-thermal-conductivity liquids. Here, it is necessary to establish, from the results of solution of the problem of heat conduction of the measurement cell, the maximum permissible values of heat fluxes and the thickness of the liquid layer, at which methodological thermal-diffusivity errors are minimum.

An analysis of the obtained results (Figs. 3 and 4) allows the conclusion on the existence of samples′ thicknesses ensuring minimum methodological errors in determining the thermal diffusivity of the liquids, all other experimental conditions being adequate. Due to the selection of the samples′ thicknesses, we can minimize the values of methodological errors in determining the thermal diffusivity (and accordingly other thermophysical characteristics) of low-thermal-conductivity liquids.

NOTATION

a, thermal diffusivity, m^2/s; c, specific heat, J/(kg·K); L, sample′s thickness, m; Q, energy absorbed by the sample, J/m^2; q, density of the radiation flux on the sample′s heated surface, W/m^2; T, temperature, K; T_0, initial value of temperature, K; T_b, boiling point of the liquid, K; T_{max} and T_{max}^{min}, maximum and minimum temperatures of superheating of the sample′s surface opposite to the heated one according to the method of [1], K; t, time, s; Δ, thickness of the liquid layer, m; δ_a, errors in determining the thermal diffusivity of the material, %; λ, thermal conductivity, W/(m·K); ν, coefficient of variation in the step of x near the boundaries of materials; ρ, density, kg/m^3; $\tau_{0.5}$, time of reaching half the value of T_{max}, s; τ_{pul}, duration of a laser-radiation pulse, s. Subscripts: 1 and 3, material of the measurement cell; 2, liquid.

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